Methods of proof – 1

The pigeonhole principle: discrete, continuous, probabilistic

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Theoretical minimum.

Fact 1 (the discrete form). If n objects are put into k boxes, some box contains at least $\lceil n/k \rceil$ of them. The whole art is choosing the boxes: residues, dyadic scales, sign patterns, first digits, positions in an ordering.

Theorem 1 (the continuous form). If sets $A_1, \dots, A_n$ inside a set of measure μ have total measure greater than $k\mu$, some point lies in at least $k+1$ of them (Blichfeldt). Two consequences: Minkowski's theorem — a centrally symmetric convex body of volume greater than 2^n contains a non-zero lattice point (Convex – 10) — and Steinhaus' theorem: if A has positive measure, then $A - A$ contains an interval around zero.

Theorem 2 (Dirichlet's approximation). For every irrational α there are infinitely many fractions with $\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^2}$; the proof is the pigeonhole applied to the fractional parts $\{q\alpha\}$ in N boxes. The same argument gives the density of $\{n\alpha\}$, which was upgraded to equidistribution in Trig-5.

Fact 2 (the probabilistic form). There is always an element at least as good as the average: if $\mathrm{E}\, X = m$ then $X \geqslant m$ with positive probability. This is the pigeonhole with weights, and the second moment refines it — a small variance means that almost every element is close to the average (Probability – 1 and Inequalities – 8).

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1. Prove the discrete pigeonhole in the form with $\lceil n/k \rceil$ and deduce Dirichlet's approximation theorem; then prove that for every irrational α and every N there is $q \leqslant N$ with $\|q\alpha\| < \frac{1}{N}$, and deduce the density of $\{n\alpha\}$ in $[0,1]$.

2. Prove Blichfeldt's lemma and derive from it both Minkowski's theorem on a symmetric convex body and Steinhaus' theorem that $A-A$ contains an interval whenever $m(A) > 0$. Deduce that an additive function bounded on a set of positive measure is linear (compare with Continuous functions – 6).

3. (IMC) Let X be a set of $\binom{2k-4}{k-2} + 1$ real numbers, $k \geqslant 2$. Prove that there exists a monotone sequence $\{x_i\}_{i=1}^{k} \subseteq X$ such that $|x_{i+1} - x_1| \geqslant 2|x_i - x_1|$ for all $i = 2, \ldots, k-1$.

4. Prove that among any $n+1$ numbers chosen from $\{1, 2, \ldots, 2n\}$ some one divides another, and that among any n^2+1 distinct reals there is a monotone subsequence of length $n+1$; identify the boxes in each case.

5. Prove by a counting argument that for $k \geqslant 3$ and $n < 2^{k/2}$ there exists a two-colouring of the edges of K_n with no monochromatic K_k, and state the general principle: to prove existence, count the bad configurations and show that they cannot cover everything.